Constructive surreal addition preserving strict and non-strict order

The Univalent Foundations Program

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Editorial note. One counted candidate: the author example eg:surreal-addition, constructing addition and proving preservation of both inequalities. The preceding complete order-code theorem is supplied author context, not a second counted target.

Editorial note. The author uses a dependent inner induction and explicitly treats the final left-order cases by symmetry. This original author compression is preserved; no generated proof completion is supplied.

Editorial note. Proposed H2 and targeted semantic screening are reviewable metadata only. This local material has no public stable ID and no publication or owner admission.

Allowed standard prerequisites (editorial).

- Basic univalent type theory and supplied surreal HIIT elimination. Univalence, function extensionality, dependent sums/products, mere proposition elimination, and supplied original surreal HIIT constructors/mutual induction. Original order-code/transitivity author proof supplied as context.

Source reference. Original theorem 11.3.16: sources/main.labelnumbers.first-edition, source lines 1177–1177. Comparison of proof organization only; the surreal proof constructs its own motives and completes all of its own cases. Cauchy approximation is not used as a theorem.

Source reference. Original theorem 7.2.2: sources/main.labelnumbers.first-edition, source lines 1797–1797. Explicitly allowed basic sethood criterion: a reflexive mere relation implying identity makes a type a set; not an order-code construction.

Definition 11.6.1. The type No of **surreal numbers**, along with the relations $< : \mathsf{No} \to \mathsf{No} \to \mathcal{U}$ and $\leq : \mathsf{No} \to \mathsf{No} \to \mathcal{U}$, are defined higher inductive-inductively as follows. The type No has the following constructors.

- For any $\mathcal{L}, \mathcal{R} : \mathcal{U}$ and functions $\mathcal{L} \to \mathsf{No}$ and $\mathcal{R} \to \mathsf{No}$, whose values we write as x^L and x^R for $L : \mathcal{L}$ and $R : \mathcal{R}$ respectively, if $\forall(L : \mathcal{L}).\, \forall(R : \mathcal{R}).\, x^L < x^R$, then there is a surreal number x.
- For any $x, y : \mathsf{No}$ such that $x \leq y$ and $y \leq x$, we have $\mathsf{eq}_{\mathsf{No}}(x, y) : x = y$.

We will refer to the inputs of the first constructor as a **cut**. If x is the surreal number constructed from a cut, then the notation x^L will implicitly assume $L : \mathcal{L}$, and similarly x^R will assume $R : \mathcal{R}$. In this way we can usually avoid naming the indexing types $\mathcal{L}$ and $\mathcal{R}$, which is convenient when there are many different cuts under discussion. Following Conway, we call x^L a *left option* of x and x^R a *right option*.

The path constructor implies that different cuts can define the same surreal number. Thus, it does not make sense to speak of the left or right options of an arbitrary surreal number x, unless we also know that x is defined by a particular cut. Thus in what follows we will say, for instance, "given a cut defining a surreal number x" in contrast to "given a surreal number x".

The relation $\leq$ has the following constructors.

- Given cuts defining two surreal numbers x and y, if $x^L < y$ for all L, and $x < y^R$ for all R, then $x \leq y$.
- Propositional truncation: for any $x, y : \mathsf{No}$, if $p, q : x \leq y$, then $p = q$.

And the relation $<$ has the following constructors.

- Given cuts defining two surreal numbers x and y, if there is an L such that $x \leq y^L$, then $x < y$.
- Given cuts defining two surreal numbers x and y, if there is an R such that $x^R \leq y$, then $x < y$.
- Propositional truncation: for any $x, y : \mathsf{No}$, if $p, q : x < y$, then $p = q$.

In order to say much more about surreal numbers, however, we need their induction principle. The mutual induction principle for $(\mathsf{No}, \leq, <)$ applies to three families of types:

$$\begin{aligned} A &: \mathsf{No} \to \mathcal{U} \\ B &: \prod_{(x,y:\mathsf{No})} \prod_{(a:A(x))} \prod_{(b:A(y))} (x \leq y) \to \mathcal{U} \\ C &: \prod_{(x,y:\mathsf{No})} \prod_{(a:A(x))} \prod_{(b:A(y))} (x < y) \to \mathcal{U}. \end{aligned}$$

As with the induction principle for Cauchy reals, it is helpful to think of B and C as families of relations between the types $A(x)$ and $A(y)$. Thus we write $B(x, y, a, b, \xi)$ as $(x, a) \trianglelefteqslant^{\xi} (y, b)$ and $C(x, y, a, b, \xi)$ as $(x, a) \triangleleft^{\xi} (y, b)$. Similarly, we usually omit the ξ since it inhabits a mere proposition and so is uninteresting, and we may often omit x and y as well, writing simply $a \trianglelefteqslant b$ or $a \triangleleft b$. With these notations, the hypotheses of the induction principle are the following.

- For any cut defining a surreal number x, together with

  (i) for each L, an element $a^L : A(x^L)$, and
  (ii) for each R, an element $a^R : A(x^R)$, such that
  (iii) for all L and R we have $(x^L, a^L) \triangleleft (x^R, a^R)$

  there is a specified element $f_a : A(x)$. We call such data a **dependent cut** over the cut defining x.
- For any $x, y : \mathsf{No}$ with $a : A(x)$ and $b : A(y)$, if $x \leq y$ and $y \leq x$ and also $(x, a) \trianglelefteqslant (y, b)$ and $(y, b) \trianglelefteqslant (x, a)$, then $a =^{A}_{\mathsf{eq}_{\mathsf{No}}} b$.

- Given cuts defining two surreal numbers x and y, and dependent cuts a over x and b over y, such that for all L we have $x^L < y$ and $(x^L, a^L) \lhd (y, f_b)$, and for all R we have $x < y^R$ and $(x, f_a) \lhd (y^R, b^R)$, then $(x, f_a) \trianglelefteq (y, f_b)$.
- $\trianglelefteq$ takes values in mere propositions.
- Given cuts defining two surreal numbers x and y, dependent cuts a over x and b over y, and an L_0 such that $x \leq y^{L_0}$ and $(x, f_a) \trianglelefteq (y^{L_0}, b^{L_0})$, we have $(x, f_a) \lhd (y, f_b)$.
- Given cuts defining two surreal numbers x and y, dependent cuts a over x and b over y, and an R_0 such that $x^{R_0} \leq y$ together with $(x^{R_0}, a^{R_0}), \trianglelefteq (y, f_b)$, we have $(x, f_a) \lhd (y, f_b)$.
- $\lhd$ takes values in mere propositions.

Under these hypotheses we deduce a function $f : \prod_{(x:\mathsf{No})} A(x)$ such that

$$\begin{aligned} f(x) &\equiv f_{f[x]} \\ (x \leq y) &\Rightarrow (x, f(x)) \trianglelefteq (y, f(y)) \\ (x < y) &\Rightarrow (x, f(x)) \lhd (y, f(y)). \end{aligned} \tag{11.6.2}$$

In the computation rule (11.6.2) for the point constructor, x is a surreal number defined by a cut, and $f[x]$ denotes the dependent cut over x defined by applying f (and using the fact that f takes $<$ to $\lhd$). As usual, we will generally use pattern-matching notation, where the definition of f on a cut $\{\, x^L \mid x^R \,\}$ may use the symbols $f(x^L)$ and $f(x^R)$ and the assumption that they form a dependent cut.

As with the Cauchy reals, we have special cases resulting from trivializing some of A, $\trianglelefteq$, and $\lhd$. Taking $\trianglelefteq$ and $\lhd$ to be constant at $\mathbf{1}$, we have **No-induction**, which for simplicity we state only for mere properties:

- Given $P : \mathsf{No} \to \mathsf{Prop}$, if $P(x)$ holds whenever x is a surreal number defined by a cut such that $P(x^L)$ and $P(x^R)$ hold for all L and R, then $P(x)$ holds for all $x : \mathsf{No}$.

Theorem 11.6.3 (Conway's Theorem 0)**.**

(i) For any $x : \mathsf{No}$, we have $x \leq x$.

(ii) For any $x : \mathsf{No}$ defined by a cut, we have $x^L < x$ and $x < x^R$ for all L and R.

Proof. Note first that if $x \leq x$, then whenever x occurs as a left option of some cut y, we have $x < y$ by the first constructor of $<$, and similarly whenever x occurs as a right option of a cut y, we have $y < x$ by the second constructor of $<$. In particular, (i)⇒(ii).

We prove (i) by No-induction on x. Thus, assume x is defined by a cut such that $x^L \leq x^L$ and $x^R \leq x^R$ for all L and R. But by our observation above, these assumptions imply $x^L < x$ and $x < x^R$ for all L and R, yielding $x \leq x$ by the constructor of $\leq$. □

Corollary 11.6.4. No *is a 0-type.*

Proof. The mere relation $R(x, y) :\equiv (x \leq y) \wedge (y \leq x)$ implies identity by the path constructor of No, and contains the diagonal by original theorem 11.6.3(i). Thus, original theorem 7.2.2 applies. □

We will also need the joint recursion principle, $(\mathsf{No}, \leq, <)$**-recursion**. It is convenient to state this as follows. Suppose A is a type equipped with relations $\trianglelefteq : A \to A \to \mathsf{Prop}$ and $\lhd : A \to A \to \mathsf{Prop}$. Then we can define $f : \mathsf{No} \to A$ by doing the following.

(i) For any x defined by a cut, assuming $f(x^L)$ and $f(x^R)$ to be defined such that $f(x^L) \lhd f(x^R)$ for all L and R, we must define $f(x)$. (We call this the *primary clause* of the recursion.)

(ii) Prove that $\trianglelefteq$ is *antisymmetric*: if $a \trianglelefteq b$ and $b \trianglelefteq a$, then $a = b$.

(iii) For x, y defined by cuts such that $x^L < y$ for all L and $x < y^R$ for all R, and assuming inductively that $f(x^L) \lhd f(y)$ for all L, $f(x) \lhd f(y^R)$ for all R, and also that $f(x^L) \lhd f(x^R)$ and $f(y^L) \lhd f(y^R)$ for all L and R, we must prove $f(x) \trianglelefteq f(y)$.

(iv) For x, y defined by cuts and an L_0 such that $x \leq y^{L_0}$, and assuming inductively that $f(x) \trianglelefteq f(y^{L_0})$, and also that $f(x^L) \lhd f(x^R)$ and $f(y^L) \lhd f(y^R)$ for all L and R, we must prove $f(x) \lhd f(y)$.

(v) For x, y defined by cuts and an R_0 such that $x^{R_0} \leq y$, and assuming inductively that $f(x^{R_0}) \trianglelefteq f(y)$, and also that $f(x^L) \lhd f(x^R)$ and $f(y^L) \lhd f(y^R)$ for all L and R, we must prove $f(x) \lhd f(y)$.

The last three clauses can be more concisely described by saying we must prove that f (as defined in the first clause) takes $\leq$ to $\trianglelefteq$ and $<$ to $\lhd$. We will refer to these properties by saying that f *preserves inequalities*. Moreover, in proving that f preserves inequalities, we may assume the particular instance of $\leq$ or $<$ to be obtained from one of its constructors, and we may also use inductive hypotheses that f preserves all inequalities appearing in the input to that constructor.

If we succeed at (i)–(v) above, then we obtain $f : \mathsf{No} \to A$, which computes on cuts as specified by (i), and which preserves all inequalities:

$$\forall(x, y : \mathsf{No}).\, \Big((x \leq y) \to (f(x) \trianglelefteq f(y))\Big) \wedge \Big((x < y) \to (f(x) \lhd f(y))\Big).$$

Like $(\mathbb{R}_{\mathsf{c}}, \sim)$-recursion for the Cauchy reals, this recursion principle is essential for defining functions on No, since we cannot first define a function on "pre-surreals" and only later prove that it respects the notion of equality.

Theorem 11.6.5. *There are relations $\preceq : \mathsf{No} \to \mathsf{No} \to \mathsf{Prop}$ and $\prec : \mathsf{No} \to \mathsf{No} \to \mathsf{Prop}$ such that if x and y are surreals defined by cuts, then*

$$(x \preceq y) :\equiv \big(\forall(L).\, x^L \prec y\big) \wedge \big(\forall(R).\, x \prec y^R\big)$$
$$(x \prec y) :\equiv \big(\exists(L).\, x \preceq y^L\big) \vee \big(\exists(R).\, x^R \preceq y\big).$$

Moreover, we have

$$(x \prec y) \to (x \preceq y) \tag{11.6.6}$$

and all the reasonable transitivity properties making $\prec$ and $\preceq$ into a "bimodule" over $\leq$ and $<$:

$$\begin{array}{ll} (x \leq y) \to (y \preceq z) \to (x \preceq z) & (x \preceq y) \to (y \leq z) \to (x \preceq z) \\ (x \leq y) \to (y \prec z) \to (x \prec z) & (x \preceq y) \to (y < z) \to (x \prec z) \\ (x < y) \to (y \preceq z) \to (x \prec z) & (x \prec y) \to (y \leq z) \to (x \prec z). \end{array} \tag{11.6.7}$$

Proof. We define $\preceq$ and $\prec$ by double $(\mathsf{No}, \leq, <)$-induction on x, y. The first induction is a simple recursion, whose codomain is the subset A of $(\mathsf{No} \to \mathsf{Prop}) \times (\mathsf{No} \to \mathsf{Prop})$ consisting of pairs of predicates of which one implies the other and which satisfy "transitivity on the right", i.e. (11.6.6) and the right column of (11.6.7) with $(x \preceq -)$ and $(x \prec -)$ replaced by the two given predicates. As in the proof of original theorem 11.3.16, we regard these predicates as half of binary relations, writing them as $y \mapsto (\diamondsuit \preceq y)$ and $y \mapsto (\diamondsuit \prec y)$, with $\diamondsuit$ denoting the pair of relations. We equip A with the following two relations:

$$(\diamondsuit \trianglelefteq \heartsuit) :\equiv \forall(y : \mathsf{No}).\, \Big((\heartsuit \preceq y) \to (\diamondsuit \preceq y)\Big) \wedge \Big((\heartsuit \prec y) \to (\diamondsuit \prec y)\Big),$$
$$(\diamondsuit \triangleleft \heartsuit) :\equiv \forall(y : \mathsf{No}).\, \Big((\heartsuit \preceq y) \to (\diamondsuit \prec y)\Big).$$

Note that $\trianglelefteq$ is antisymmetric, since if $\diamondsuit \trianglelefteq \heartsuit$ and $\heartsuit \trianglelefteq \diamondsuit$, then $(\heartsuit \preceq y) \Leftrightarrow (\diamondsuit \preceq y)$ and $(\heartsuit \prec y) \Leftrightarrow (\diamondsuit \prec y)$ for all y, hence $\diamondsuit = \heartsuit$ by univalence for mere propositions and function extensionality. Moreover, to say that a function $\mathsf{No} \to A$ preserves inequalities is exactly to say that, when regarded as a pair of binary relations on No, it satisfies "transitivity on the left" (the left column of (11.6.7)).

Now for the primary clause of the recursion, we assume given x defined by a cut, and relations $(x^L \prec -)$, $(x^R \prec -)$, $(x^L \preceq -)$, and $(x^R \preceq -)$ for all L and R, of which the strict ones imply the non-strict ones, which satisfy transitivity on the right, and such that

$$\forall(L, R).\, \forall(y : \mathsf{No}).\, \Big((x^R \preceq y) \to (x^L \prec y)\Big). \tag{11.6.8}$$

We now have to define $(x \prec y)$ and $(x \preceq y)$ for all y. Here in contrast to original theorem 11.3.16, rather than a nested recursion, we use a nested induction, in order to be able to inductively use transitivity on the left with respect to the inequalities $x^L < x$ and $x < x^R$. Define $A' : \mathsf{No} \to \mathcal{U}$ by taking $A'(y)$ to be the subset A' of $\mathsf{Prop} \times \mathsf{Prop}$ consisting of two mere propositions, denoted $\triangle \preceq y$ and $\triangle \prec y$ (with $\triangle : A'(y)$), such that

$$(\triangle \prec y) \to (\triangle \preceq y) \tag{11.6.9}$$
$$\forall(L).\, (\triangle \preceq y) \to (x^L \prec y) \tag{11.6.10}$$
$$\forall(R).\, (x^R \preceq y) \to (\triangle \prec y). \tag{11.6.11}$$

Using notation analogous to $\trianglelefteq$ and $\triangleleft$, we equip A' with the two relations defined for $\triangle : A'(y)$ and $\square : A'(z)$ by

$$(\triangle \sqsubseteq \square) :\equiv \Big((\triangle \preceq y) \to (\square \preceq z)\Big) \wedge \Big((\triangle \prec y) \to (\square \prec z)\Big)$$
$$(\triangle \sqsubset \square) :\equiv \Big((\triangle \preceq y) \to (\square \prec z)\Big).$$

Again, $\sqsubseteq$ is evidently antisymmetric in the appropriate sense. Moreover, a function $\prod_{(y:\mathsf{No})} A'(y)$ which preserves inequalities is precisely a pair of predicates of which one implies the other, which satisfy transitivity on the right, and transitivity on the left with respect to the inequalities $x^L < x$ and $x < x^R$. Thus, this inner induction will provide what we need to complete the primary clause of the outer recursion.

For the primary clause of the inner induction, we assume also given y defined by a cut, and properties $(x \prec y^L)$, $(x \prec y^R)$, $(x \preceq y^L)$, and $(x \preceq y^R)$ for all L and R, with the strict ones implying the non-strict ones, transitivity on the left with respect to $x^L < x$ and $x < x^R$, and on the right with respect to $y^L < y^R$. We can now give the definitions specified in the theorem statement:

$$(x \preceq y) :\equiv (\forall(L).\, x^L \prec y) \wedge (\forall(R).\, x \prec y^R), \tag{11.6.12}$$

$$(x \prec y) :\equiv (\exists(L).\, x \preceq y^L) \vee (\exists(R).\, x^R \preceq y). \tag{11.6.13}$$

For this to define an element of $A'(y)$, we must show first that $(x \prec y) \to (x \preceq y)$. The assumption $x \prec y$ has two cases. On one hand, if there is L_0 with $x \preceq y^{L_0}$, then by transitivity on the right with respect to $y^{L_0} < y^R$, we have $x \prec y^R$ for all R. Moreover, by transitivity on the left with respect to $x^L < x$, we have $x^L \prec y^{L_0}$ for any L, hence $x^L \prec y$ by transitivity on the right. Thus, $x \preceq y$.

On the other hand, if there is R_0 with $x^{R_0} \preceq y$, then by (11.6.8), we have $x^L \prec y$ for all L. And by transitivity on the left and right with respect to $x < x^{R_0}$ and $y < y^R$, we have $x \prec y^R$ for any R. Thus, $x \preceq y$.

We also need to show that these definitions are transitive on the left with respect to $x^L < x$ and $x < x^R$. But if $x \preceq y$, then $x^L \prec y$ for all L by definition; while if $x^R \preceq y$, then $x \prec y$ also by definition.

Thus, (11.6.12) and (11.6.13) do define an element of $A'(y)$. We now have to verify that this definition preserves inequalities, as a dependent function into A', i.e. that these relations are transitive on the right. Remember that in each case, we may assume inductively that they are transitive on the right with respect to all inequalities arising in the inequality constructor.

- Suppose $x \preceq y$ and $y \leq z$, the latter arising from $y^L < z$ and $y < z^R$ for all L and R. Then the inductive hypothesis (of the inner recursion) applied to $y < z^R$ yields $x \prec z^R$ for any R. Moreover, by definition $x \preceq y$ implies that $x^L \prec y$ for any L, so by the inductive hypothesis of the outer recursion we have $x^L \prec z$. Thus, $x \preceq z$.
- Suppose $x \preceq y$ and $y < z$. First, suppose $y < z$ arises from $y \leq z^{L_0}$. Then the inner inductive hypothesis applied to $y \leq z^{L_0}$ yields $x \preceq z^{L_0}$, hence $x \prec z$.

  Second, suppose $y < z$ arises from $y^{R_0} \leq z$. Then by definition, $x \preceq y$ implies $x \prec y^{R_0}$, and then the inner inductive hypothesis for $y^{R_0} \leq z$ yields $x \prec z$.
- Suppose $x \prec y$ and $y \leq z$, the latter arising from $y^L < z$ and $y < z^R$ for all L and R. By definition, $x \prec y$ implies there merely exists R_0 with $x^{R_0} \preceq y$ or L_0 with $x \preceq y^{L_0}$. If $x^{R_0} \preceq y$, then the outer inductive hypothesis yields $x^{R_0} \preceq z$, hence $x \prec z$. If $x \preceq y^{L_0}$, then the inner inductive hypothesis for $y^{L_0} < z$ (which holds by the constructor of $y \leq z$) yields $x \prec z$.

This completes the inner induction. Thus, for any x defined by a cut, we have $(x \prec -)$ and $(x \preceq -)$ defined by (11.6.12) and (11.6.13), and transitive on the right.

To complete the outer recursion, we need to verify these definitions are transitive on the left. After a No-induction on z, we end up with three cases that are essentially identical to those just described above for transitivity on the right. Hence, we omit them. □

Theorem 11.6.14. *For any $x, y : \mathsf{No}$ we have $(x < y) = (x \prec y)$ and $(x \leq y) = (x \preceq y)$.*

Proof. From left to right, we use $(\mathsf{No}, \leq, <)$-induction where $A(x) :\equiv \mathbf{1}$, with $\preceq$ and $\prec$ supplying the relations $\trianglelefteq$ and $\triangleleft$. In all the constructor cases, x and y are defined by cuts, so the definitions of $\preceq$ and $\prec$ evaluate, and the inductive hypotheses apply.

From right to left, we use No-induction to assume that x and y are defined by cuts. But now the definitions of $\preceq$ and $\prec$, and the inductive hypotheses, supply exactly the data required for the relevant constructors of $\leq$ and $<$. □

Corollary 11.6.15. *The relations $\leq$ and $<$ on No satisfy*

$$\forall(x, y : \mathsf{No}).\, (x < y) \to (x \leq y)$$

and are transitive:

$$(x \leq y) \to (y \leq z) \to (x \leq z)$$
$$(x \leq y) \to (y < z) \to (x < z)$$
$$(x < y) \to (y \leq z) \to (x < z).$$

As with the Cauchy reals, the joint $(\mathsf{No}, \leq, <)$-recursion principle remains essential when defining all operations on No.

Example 11.6.16. We define $+ : \mathsf{No} \to \mathsf{No} \to \mathsf{No}$ by recursion on the first argument, followed by induction on the second argument. For the outer recursion, we take the codomain to be the subset of $\mathsf{No} \to \mathsf{No}$ consisting of functions g such that $(x < y) \to (g(x) < g(y))$ and $(x \leq y) \to (g(x) \leq g(y))$ for all x, y. For such g, h we define $(g \trianglelefteq h) :\equiv \forall(x : \mathsf{No}).\, g(x) \leq h(x)$ and $(g \triangleleft h) :\equiv \forall(x : \mathsf{No}).\, g(x) < h(x)$. Clearly $\trianglelefteq$ is antisymmetric.

For the primary clause of the recursion, we suppose x defined by a cut, that the functions $(x^L + -)$ and $(x^R + -)$ are defined, preserve inequalities, and satisfy $x^L + y < x^R + y$, and we define $(x + -)$. As in original theorem 11.6.5, rather than an inner recursion, we use an inner induction into the family $A : \mathsf{No} \to \mathcal{U}$, where $A(y)$ is the subset of those $z : \mathsf{No}$ such that each $x^L + y < z$ and each $x^R + y > z$. We equip A with the relations $\leq$ and $<$ induced from No, so that antisymmetry is obvious. For the primary clause of the inner recursion, we suppose also y defined by a cut, with each $x + y^L$ and $x + y^R$ defined and satisfying $x^L + y^L < x + y^L$, $x^L + y^R < x + y^R$, $x + y^L < x^R + y^L$, and $x + y^R < x^R + y^R$ (these come from the additional conditions imposed on elements of $A(y)$), and also $x + y^L < x + y^R$ (since the elements $x + y^L$ and $x + y^R$ of $A(y)$ form a dependent cut). Now we give Conway's definition:

$$x + y :\equiv \{\, x^L + y, x + y^L \,\big|\, x^R + y, x + y^R \,\}.$$

In other words, the left options of $x + y$ are all numbers of the form $x^L + y$ for some left option x^L, or $x + y^L$ for some left option y^L. We must show that each of these left options is less than each of these right options:

- $x^L + y < x^R + y$ by the outer inductive hypothesis.
- $x^L + y < x^L + y^R < x + y^R$, the first since $(x^L + -)$ preserves inequalities, and the second since $x + y^R : A(y^R)$.

- $x + y^L < x^R + y^L < x^R + y$, the first since $x + y^L : A(y^L)$ and the second since $(x^R + -)$ preserves inequalities.
- $x + y^L < x + y^R$ by the inner inductive hypothesis (specifically, the fact that we have a dependent cut).

We also have to show that $x+y$ thusly defined lies in $A(y)$, i.e. that $x^L+y < x+y$ and $x + y < x^R + y$; but this is true by original theorem 11.6.3(ii).

Next we have to verify that the definition of $(x + -)$ preserves inequality:

- If $y \leq z$ arises from knowing that $y^L < z$ and $y < z^R$ for all L and R, then the inner inductive hypothesis gives $x+y^L < x+z$ and $x+y < x+z^R$, while the outer inductive hypotheses give $x^L + y \leq x^L + z$ and $x^R + y \leq x^R + z$. Moreover, since $x^R + y$ is by definition a right option of $x + y$, we have $x + y < x^R + y$. Similarly, we find that $x^L + z$ is a left option of $x + z$, so that $x^L + z < x + z$. Thus, using transitivity, we have $x^L + y < x + z$ and $x + y < x^R + z$; so we may conclude $x + y \leq x + z$ by the constructor of $\leq$.
- If $y < z$ arises from an L_0 with $y \leq z^{L_0}$, then inductively $x + y \leq x + z^{L_0}$, hence $x + y < x + z$ since $x + z^{L_0}$ is a right option of $x + z$.
- Similarly, if $y < z$ arises from $y^{R_0} \leq z$, then $x + y < x + z$ since $x + y^{R_0} \leq x + z$.

This completes the inner induction. For the outer recursion, we have to verify that + preserves inequality on the left as well. After an No-induction, this proceeds in exactly the same way.